AI x Crypto ~ Active Entity Ontology for Science & Active Blockference ~ Active Inference Institute

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hi everyone I'm jakub and uh today I'll um I'm going to give a very brief overview of uh some projects that we've been developing at the Active inference Institute and their relevance to to decentralized AI now I think my uh talk and Alex is right after me is going to overlap slightly so uh I'm going to try to keep the theory of active inference and one particular project to um a minimum although I'll try to give some brief um brief introduction to what active inference is so that you uh have the relevant background so since this is a mini conference on decentralized AI I'm going to assume that um most of you have at least heard of reinforcement learning which on the computational side is quite similar to active inference but we can think of active inference as being a kind of generalization of reinforcement learning where agents uh interact with their environment or and try to solve a certain task or try to navigate in their environment but instead of um maximizing some utility function or uh instead of their goal being always maximizing for reward um active inference agents are perhaps more General where the kind of utility function or the general function of active inference agents is always uncertainty minimization so active inference agents form some kind of internal model of their environment and they can learn more about their environment by engaging with it through action and then um observing what uh what kind of uh sense sensory input goes from the environment to the agents so there's a kind of Duality between perception and action that's always mediated through uncertainty minimization and uh because of its generality this framework can be applied to pretty much any system that's um Salient enough to persist through time which was kind of the motivation for uh the first project which Alex will talk about a bit more so I'm only going to mention it briefly which is the active entity ontology for science uh which really stands at the intersection of three Fields namely DSi systems engineering and active inference where our aim was to create an ontology that individuals and communities within DSi can use to collectively minimize their uncertainty about what uh kind of system they find themselves in and how that system might evolve through time and I think this is uh the same can be done for decentralized AI which has a lot of uh similar artifacts that have to do with web3 and um perhaps we can say that either DSi is encapsulated within decentralized AI or the other way around depending on uh which you think is the bigger uh bigger set but the core the core utility of having a shared ontology that's kind of grounded in another formal framework namely active inference allows us to more um specifically Define what kind of entities and artifacts we can find in a system and how um how these entities interact with each other and minimize their minimize their uncertainty uh so we can kind of have a better understanding of the Dynamics of our system and be able to better um make decisions about what kind of protocols we should create or how should our uh community interact with other communities and uh and so on this also uh leads me to uh the active blog friends project which aims which is kind of on the computational side of EOS where we aim to uh combine um the open source framework CAD CAD which was developed by block science and that's been used uh mainly for token engineering and also just generalized dynamical systems modeling um where you usually take a kind of structural top-down approach to modeling uh your system of inner interest where you define your state space and then you perform giant parameter sweeps to try to figure out uh how your state space evolves through time and what kind of edge cases can you um can you get to and combining that with active inference cognitive modeling where you have agents that have kind of emergent properties that through Collective infra interaction form these uh communities that themselves can be can be cast as agents within some environment so you have this um bottom uh bottom-up emergent Behavior through agent agents interacting and trying to minimize their uncertainty as well as um a kind of top-down parameter sweep over all the possible States in your system um this might sound a little abstract but I think um it's quite relevant for decentralized AI and mainly for two different reasons one of them is we uh we want to create different communities and protocols for both uh co-owning or creating AI models or interacting with AI models and that's already a quite um quite large State space so it would be help it would definitely be about AI alignment which is also a very big question in a big problem in in AI where you're basically trying to align uncertainty minimization across different types of Agents as well as um different levels of your system where doubt want to minimize their uncertainty with respect to other doubts or with respect to um some uh some preset goal but individuals within those doubts also need to uh minimize their uncertainty based on their own preferences which may or may not align with those of of the higher level system and the same can be said of um of AI models and autonomous agents which are deaf which are definitely becoming more um more present in the in the digital space so that would be a very uh brief overview of some of the projects that were developing at the Active inference Institute and uh if you have any questions then I'm not sure whether we have the time but thanks a lot for your for your attention yeah you have a couple you have about a minute so um maybe we'll save them for a little bit later unless they're quite quick okay I'm not seeing any text questions yet so um I'll definitely invite people to ask you any questions that they have about that either in the 10 minute session or towards the end when we finish um we've got about a minute now so I will thank you very much for that very cool to hear what you're doing I'll bring up your your colleague uh now hi Alex how are you yeah fine thanks hi everyone cool take it away okay can you see it right so I will add here some information about active inference and a a a EOS and continue to Jacob talk but also will provide more information from Institute scale about our work and possible collaboration service our participants so we are Institute which is working on Frontier or effective inference as we consider it and it's a scale free cognitive framework which is based based on first principles but models decision making processes and actions of actions in the environment and it's quite important about scale free here because it means that we can apply it for different types of Agents and systems and for me personally very important that which but it's based on first principles starting from from physics from thermodynamics from information Theory to Quantum information Theory it's quite broad space for people who are like math and this framework combines perception actions and inference inference optimize information processing and handle uncertainty in relation to different kinds of Agents it can be polite which was mentioned for robotics for artificial intelligence but also a lot our fields and domains like neuroscience and with for now we see real world examples of application of the framework active inference agent acting to minimize his surprises in informational sense and maximize maximize expected outcomes on different time scales having his generative model in place and updating it or updating environment uh this process always based on gathering information and based on it generating actions based on that generative probabilistic models as I mentioned it could be applied for various fields and we are as Institute starting it from ourselves in the beginning from Team Communications and later receive Buffs to decision making management governance and a lot of other things related to organizations as entities and as as agents as for active entity ontology for science first of all we could see in terms of active inference we can consider it as a part of a shared General generative model of people's personalities or organizations or projects which should have some common language in place and having it there are some Evolution happening and providing ways to structure to represent different scientific knowledges in a common way to enable efficient collaborations and knowledge sharing it can facilitate interoperability across different scientific domains standardized concept and relationships and finally it could be a way to accelerate scientific discoveries and Inter inter interdisciplinary collaborations you can find some additional information and to actually paper by with links and the ontology as itself you can find on coder page maybe have a look and consider if it's connects with you in some way and important here that how it could be developed both active inference as a framework as a Cornerstone and in developments with active inference can improve decision making and collaboration of teams of organizations there are possible ways and different ways to integrate it with blockchain standards which enhanced digitalizations of such scientific artifacts which later in evolutionary manner could be reproduced and evolved for all different communities finally we consider Institute as a part of research ecosystem and our intentions here to provide we as an agent to provide more interfaces and more structured ways to communicate in this Frontier domains because it's quite hard science and we need to find ways to use common language in different projects and by that of all evolve research existence so I think that's all from from my site so if you have some questions can you hear me nice way yeah we can we'll give it a few minutes maybe if people want to type some questions okay really cool stuff it's incredible how complex and I can see a lot of different ways that you could collaborate with people in many many different fields so it's very cool cool thank you